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RESEARCH ARTICLE

An Enhanced Energy Optimization Model for Industrial Wireless Sensor Networks Using Machine Learning

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ABSTRACT Industrial Wireless Sensor Networks (WSNs) are becoming increasingly popular due to their enhanced scalability and low cost of deployment. However, they also present new challenges, such as energy optimization and network maintenance, which industrial users must address. In order to meet the challenges, Machine Learning techniques have been used to create an enhanced energy optimization model for Industrial WSNs. This model utilizes knowledge-based learning to identify and optimize the energy consumption of the nodes, allowing Industrial WSNs to consume the least amount of energy for the given tasks. In addition, the model also evaluates the effectiveness of feedback control schemes and predicts the best possible outcomes for its application in Industrial WSNs to ensure higher efficiency and longer network lifetime. The model also enables the exploration of potential trade-offs between power consumption and communication performance to ensure a better energy-efficient solution. The proposed EEOM obtained 64.72% transmission energy consumption, 35.28% transmission energy saving, 67.27% received energy consumption, 32.73% received energy storage, 52.16% idle-mode energy consumption, 47.84% idle-mode energy storage, 66.31% sleep-mode energy consumption, and 33.69% sleep-mode energy storage. It also obtained 90.44% prevalence threshold, 90.33% critical success index, 93.93% Delta-P, 90.06% MCC and 92.17% FMI rates. It also provides the ability to identify the best selection of nodes and paths for data transmission to reduce network traffic. When applied in conjunction with manual intervention, these automated knowledge-based techniques will make Industrial WSNs more reliable, efficient, and energy-cost effective.

INDEX TERMS Industrial, wireless, sensor, networks, WSN, scalability, machine learning, efficient, energy.

I. INTRODUCTION

Industrial wireless sensor networks (IWSNs) are specialized wireless networks used in industrial settings, such as industrial automation, process control, monitoring, and safety [1]. IWSNs consist of a collection of sensor nodes deployed over an area of interest for monitoring purposes [2]. They are connected to a gateway node and base station via a wireless medium to relay data to a supervisory agent or control application. Sensor nodes are composed of sensors, transceivers, microcontrollers, and power supplies and often employ multiple wireless protocols to optimize their behavior and power consumption [3]. IWSNs can monitor various physical and environmental phenomena, such as temperature, pressure,

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sound, vibration, etc [4]. They can also provide valuable information on the health and performance of complex industrial systems, helping to identify operational inefficiencies or hazardous situations. It is used to monitor and control physical processes and monitoring remote locations [5]. IWSNs allow users to access, monitor, and control various industrial resources, from environmental and climactic conditions to manufacturing processes and equipment [6]. Typical uses of IWSNS include manufacturing and process management, hazardous material detection, energy distribution and usage monitoring, and security monitoring [7].

The primary functions of an IWSN are to provide reliable and secure data between sensor nodes and the applications that need them. Sensor nodes are specialized wireless devices with the necessary hardware and software to collect, analyze, and transmit data to other nodes [8]. This data can then be used to control and manage the physical processes or locations monitored by the IWSN. Sensor nodes can also detect changes in their environment and alert other nodes, allowing for quick and efficient responses to changing conditions or events [9]. Another essential function of an IWSN is data management and analysis. IWSNs must be able to efficiently organize, store, and transport data for users to access and take advantage of the network's capabilities [10]. This data can be used for predictive analysis and forecasting, allowing users to detect trends and make better decisions. Finally, IWSNs must be reliable and secure. IWSNs must have robust communication protocols and authentication to ensure data integrity and prevent unauthorized access and tampering [11]. Additionally, IWSNs must have reliable network infrastructure components such as routers, firewalls, and other measures to provide the reliability and security necessary for an industrial application [12]. Industrial Wireless Sensor Networks (IWSNs) are complex networks in which wireless sensor nodes are used to bridge the gap between machine-level and sensor-level transactions. They can enable a new industrial automation and control era but pose several challenges [13]. There are significant technical issues when integrating the IWSNs with existing industrial systems. These include issues with network coverage, scalability, reliability, and the interoperability and security of data transmissions [14]. There are several complications and issues related to the deployment and management of these networks [15]. These include issues with the logistics of setting up the wireless infrastructure and potential risks to the safety and security of the production environment. The cost of deploying and maintaining an IWSN can hinder its adoption [16]. Companies must weigh the benefits of integrating wireless technology with their existing industrial systems against the costs associated with installation, maintenance, and support [17].

The IWSNs can offer many advantages to manufacturing industries, but these come with several complications that must be addressed. It includes issues related to technology, deployment, and cost [18]. Companies must weigh the benefits against the technical and financial implications of deploying an IWSN before making an informed decision. IWSNs are networks of wireless sensor nodes specifically designed and deployed in industrial environments [19]. The nodes in these networks are equipped with different environmental sensors ranging from temperature and humidity to chemical and motion sensors. The primary purpose of IWSNs is to demonstrate that sensing devices can be deployed in remote industrial locations at a meager cost. IWSNs offer several advantages over traditional wired sensor networks, such as ease of installation, flexibility, scalability, and fault tolerance [20]. The IWSNs can provide real-time monitoring data, which can be used for optimizing operations and increasing productivity. The latest innovations in IWSN technology are highly energy-efficient, with nodes consuming meager power. The recent developments in low-power, mesh network communication protocols have enabled an improved degree of coverage and flexibility in deployment [21]. The self-healing protocols have been developed to recover from packet loss and other communication errors automatically. Some of the latest advances in IWSN technology involve robust localization techniques and distributed target-tracking algorithms to enable multiple nodes to track a single target. Combining these techniques makes IWSNs an attractive solution for various industrial applications. The main contribution of the research has the following,

- Enhanced Productivity: Machine learning algorithms enable industrial wireless sensor networks to accurately monitor and analyze data to identify potential issues and develop new and better ways to increase production efficiency.
- Predictive Maintenance: Machine learning algorithms can detect anomalies in the data collected from industrial wireless sensor networks and predict when maintenance is needed. It helps to reduce downtime and reduce costs associated with unplanned repairs.
- Improved Safety: Machine learning algorithms can detect potential hazards in operational environments and alert operators to take corrective measures to prevent accidents.
- Automation: Industrial wireless sensor networks can automate processes and tasks, reducing labor costs and increasing accuracy.
- Improved Quality: Machine learning algorithms can detect and identify product defects before they are shipped, reducing the cost of returns and increasing customer satisfaction.

The rest of the paper has organized as the following. Section II illustrates the recent works related to the research. Section III provides the proposed model, and section IV expresses the analytical discussion. Section V shows the comparative analysis between the existing and proposed models. Finally, section VI expresses the conclusion and future scope of the research

II. LITERATURE REVIEW

Ramesh et al. [22] has discussed the efficiency of industrial wireless sensor networks (WSNs) has always been a significant area of research and development for engineers. It is because WSNs are commonly used in industrial applications involving remote monitoring and automation. As the number of WSNs deployed in various industrial applications increases, so does the need for adequate energy optimization of these networks. It is well-known that energy optimization is required for WSNs in order to maintain optimal communication performance and ensure efficient operation. Ahmad et al. [23] has discussed the advanced energy optimization models, such as the Energy Efficient Optimization (EEO), developed for industrial WSNs. The EEO model is based on the concept of using machine learning algorithms to predict node energy levels, which allows for more accurate energy utilization. The EEO can provide improved energy optimization of industrial WSNs by predictive optimization techniques and dynamic adjustment of resource-allocation strategies. Shanmugasundaram and Madheswaran et al. [24] has discussed the EEO model, energy optimization of industrial wireless sensor networks can be achieved in several ways. First, it is possible to propose algorithms using machine learning techniques to predict the nodes' energy levels accurately. It will allow the network to better allocate resources to nodes with higher remaining energy levels and reduce energy consumption overall. Matlou and Abu-Mahfouz et al. [25] has discussed monitoring energy levels across the entire network through machine learning algorithms and identified areas that require improvements that can be targeted more efficiently. The EEO includes a dynamic policy switch facility to adjust the resource-allocation strategies for different network conditions. By using this feature, it is possible to customize the resource-allocation policy based on the current network condition. Moreover, this feature allows for a fast response when network conditions change to minimize energy consumption. The EEO can be used to control the routing behavior of WSNs. When combined with machine learning, the EEO can accurately identify and optimize the routing paths, leading to efficient data routing and improved energy utilization. Praveen et al. [26] has discussed using the enhanced energy optimization model, which is based on machine learning, and can help improve the energy optimization of industrial WSNs. By using machine learning algorithms to predict node energy levels and allowing for dynamic resource allocation strategies, the EEO can provide a more efficient approach to energy optimization. The ability to control the data routing process using this model can also improve energy utilization. Yang et al. [27] has discussed the modern industrial settings, wireless sensor networks (WSNs) are increasingly being used to provide efficient and accurate control and monitoring of complex processes and systems. Various optimization strategies can further increase the networks' reliability, efficiency, and accuracy. Blake et al. [28] has discussed the Energy optimization of WSNs as a strategy that seeks to minimize energy consumption while maximizing the network's performance. This paper compares an enhanced energy optimization model for industrial wireless sensor networks (IWSNs) that employ machine learning (ML) techniques. First, we discuss existing work on the energy optimization of WSNs and describe how ML can be used to improve the performance of these models.

Galbraith and Podhorska et al. [29] has discussed a supervised learning approach for IWSNs to optimize energy consumption. Using this approach, ML algorithms, such as decision trees and artificial neural networks, can be used to identify potential problem areas, such as unbalanced node load and nodes' received signal strength (RSS), to improve energy consumption. The proposed scheme is evaluated using both simulation and experimental results. Matusowsky et al. [30] has discussed experiments performed in an indoor environment using a real-world IWSN deployment. The simulation results indicate that the proposed approach can effectively reduce energy consumption in IWSNs. The experiments show that the proposed model can significantly reduce the network's energy consumption under various varying radio channel conditions. Pinto et al. [31] has discussed the results of experiments, including the implications of the proposed energy optimization model. It also provides a set of best practices for implementing enhanced energy optimization models in industrial wireless sensor networks. By following these best practices, organizations can achieve improved network performance and more reliable operation in IWSNs. The performance-enhanced energy optimization model for industrial wireless sensor networks using machine learning is a novel approach to energy conservation in sensor networks. The model has two main components: a reinforcement learning-based energy optimization algorithm and a genetic-based optimization algorithm. Gupta et al. [32] has discussed that the reinforcement learning (RL) component focuses on adapting the sensor's communication and sensing capabilities dynamically to minimize energy consumption in the network. It enables the sensors to identify and accurately adjust their transmission powers, rates, and sensing durations, thereby ensuring the best energy-saving strategies. The genetic algorithm component, on the other hand, is responsible for the overall energy optimization of the network. It predicts the most energy-efficient data acquisition and routing paths while considering the reliability of data links, network topology, and node type. Raza and Nguyen et al. [33] has discussed that the combined efforts of both components enable the network to assist in more efficient energy utilization while maintaining its operational robustness. This enhanced energy optimization model can provide significant benefits to industrial applications. By taking into account the application's specific needs, it can suggest the most cost-effective data routing and communication decisions, resulting in minimized energy costs. Messaoud et al. [34] has discussed the help of predictive machine learning algorithms; the system can adapt to changing traffic patterns, which results in more accurate decision-making and better performance. Mahfouz et al. [35] has discussed the potential applications, including industrial safety systems; the system can be better prepared for anomalous scenarios and emergencies. Utilizing advanced machine learning techniques, this performance-enhanced energy optimization model for industrial wireless sensor networks can be an invaluable tool for various industrial applications. Kumar et al. [36] has discussed the potential to provide sensor networks with the flexibility, scalability, and energy conservation necessary to address the specific requirements of each system. This model can create a more efficient and reliable sensor network with proper implementation, increasing their effectiveness in several domains. Zhang et al. [37] have discussed online spatiotemporal modeling for robust and lightweight device-free localization in nonstationary environments. This technology enables seamless and real-time tracking of objects and people via wireless signal sensing without additional hardware or physical infrastructure. It is a handy tool in difficult-to-map or dynamic environments, as it eliminates the need to manually re-calibrate hardware or deploy large amounts of infrastructure for each new setup. In essence, it uses wireless signals to detect and track the movement of objects and people in an area. Then it creates a 3D environment model that can be used to accurately and quickly localize the person or object being tracked. Zhang et al. [38] Hierarchical sensing for robust-in-the-air handwriting recognition with commodity WiFi devices is a technique that uses multiple sensors (such as cameras or infrared cameras) and signal processing algorithms to improve the accuracy of handwriting recognition. The technique employs multiple layers of sensors that work together to capture the handwriting signal. Each layer responds to different aspects of handwriting, and the combination and data fusion of signals from the different layers allows for more robust handwriting recognition. This type of sensing technology can be used with commodity WiFi devices to recognize handwritten characters more accurately than a single-sensor system.

From the above comprehensive analysis, the following issues were identified and listed in table.1

The identified issues are,

- Interference: Node communication in industrial wireless sensor networks can be affected by interference from other wireless devices such as cell phones, Wi-Fi networks, etc.
- Limited Bandwidth: Industrial wireless sensor networks have limited bandwidth which can lead to slow data transmission and congestion.
- Security: Industrial wireless sensor networks are vulnerable to security threats such as malicious attacks, data interception, and denial-of-service attacks.
- Power Consumption: Industrial wireless sensor networks must transmit data efficiently while consuming minimal power.

Industrial wireless sensor networks (WSNs) with machine learning offer a novel approach to monitoring and controlling industrial processes. Machine learning techniques detect anomalies, identify trends, and make predictions. By leveraging the power of WSNs combined with machine learning, industrial processes can be monitored more accurately and efficientlyThis combination of WSNs and machine learning presents an exciting opportunity for the industrial sector.

The novelty of the proposed research is the ability to use ML algorithms to enable a distributed network of sensors to learn and adapt to environmental changes. It allows for more efficient and effective communication between nodes and improved data accuracy and analysis. ML algorithms can also detect anomalies in the environment, which can be used to improve the network's performance. Additionally, ML can be used to optimize the power consumption of the nodes, which can significantly reduce the network's cost.

III. PROPOSED MODEL

As our economic and energy systems become increasingly interconnected, there is a growing demand for more efficient energy models. It has brought about an increased need for industrial wireless sensor networks. These networks allow for real-time and accurate data collection and analysis using machine learning algorithms. The industrial wireless sensor network structure has shown in the following Fig.1.

This technology can be used to help optimize the efficiency of the energy system. Implementing an efficient model is beneficial in many ways. First, it can reduce the energy cost by adjusting the network parameters in real-time based on feedback. Secondly, it can increase the accuracy of the data gathered by the sensors and allow for more accurate trend analysis and prediction. Finally, it can enhance the safety and security of the environment by monitoring the environment and alerting the user to any changes that may cause a safety issue. The energy optimization model comprises an energy management layer and a machine learning algorithm. The energy management layer allows for collecting and storing data from various sensors throughout the network. Furthermore, it allows for preparing the data for analysis and adjusting control parameters according to the observed data. The machine learning algorithm takes the accumulated data, processes it, and makes predictions and recommendations to optimize the energy system's efficiency. In order to ensure that this model is effective in increasing energy efficiency, it is essential to develop algorithms that enable the system to predict energy utilization accurately and to choose the appropriate control parameters for maximum energy savings. Furthermore, these algorithms need to account for the different types of data and devices in the environment and their interactions. The implementation of this model will require sufficient network resources, as well as software and hardware, to properly monitor and analyze the data. The energy optimization model could be further developed in several ways:

- An accurate data-gathering could be employed to understand better the energy usage patterns of a particular area or process. It could include using sensors to measure

TABLE 1. Comprehensive analysis.

Author	Model	Interference Management	Bandwidth Utilization	Network Security	Energy Utilization	Power Consumption
Ramesh, G., et al. [22]	AUTO-Encoder Based Neural-Network	High	High	Limited	High	Limited
Ahmad, R., et al. [23]	Machine learning for WSN security	Moderate	Limited	High	Low	High
Shanmugasundaram, R. N., et al. [24]	Dynamic cluster head selection algorithm	Very high	Very high	High	Low	Limited
Matlou, O. G., et al. [25]	AI in software defined WSN	Low	Low	Limited	Low	High
Praveen, S. P., et al. [26]	Virtual Private Network Flow Detection	Moderate	High	Low	Moderate	Limited
Yang, L., et al. [27]	Generative Adversarial Learning for IWSN	Low	Limited	High	Low	Very High
Blake, R., et al. [28]	Sustainable CP manufacturing systems	Very high	Low	Very Low	Very High	Low
Galbraith, A., et al. [29]	Cyber-physical smart manufacturing	Moderate	Very Low	Limited	Very Low	Very High
Matusowsky, M., et al. [30]	Machine learning-based virtual sensor	Low	Limited	Very Low	Very Low	Low
Pinto, A. R., et al. [31]	Genetic machine learning algorithms	High	Very Low	Low	Moderate	Limited
Gupta, N., et al. [32]	Structured hypercube network model	Very Low	Low	Limited	Very High	Very High
Raza, M., et al. [33]	Industrial wireless sensor networks	Moderate	Limited	Very Low	Very Low	Limited
Messaoud, S., et al.	Deep federated learning	Very Low	Very Low	Low	Moderate	Low
[34]	approach					
Mahfouz, S., et al. [35]	Target tracking	Very high	High	Limited	Low	Limited
Kumar, D. P., et al. [36]	Machine learning algorithms	Moderate	High	High	Moderate	Very High
Zhang J. et al. [37]	Online spatiotemporal modeling	Limited	Low	High	High	Low
Zhang J. et al. [38]	Hierarchical sensing	High	Limited	Very High	High	Low

specific energy usage factors or deploying sophisticated analytics to uncover significant trends.

- The data analysis could be utilized to identify better areas of optimization, such as machine learning models, predictive analytics, and optimization algorithms.
- The model could be more flexible by incorporating dynamic constraints and economic considerations into the optimization calculations.

It would make the model more accurate and efficient at identifying the best energy optimization options.

A. LAYER CONFIGURATION

It is essential to have an experienced team to design the network, develop the software, and configure the hardware. Additionally, appropriate training should be given to operators and employees so that they understand the system and how it functions. Implementing an efficient model can drastically improve industrial sites' efficiency worldwide. This technology can save energy, reduce costs, and improve safety and security. As such, it is worth further examining as a viable solution for industrial sites. This model is a new and powerful technology that allows communication between multiple nodes more efficiently and robustly. This technology can be used to design intelligent digital networks for various industrial applications, such as monitoring, industrial automation, and digital control of industrial devices. Layer configuration as shown in the Fig.2. In these networks, the nodes are embedded solar nodes equipped with hardware and software that allows for data collection and communication with the other nodes in the network. The solar nodes use

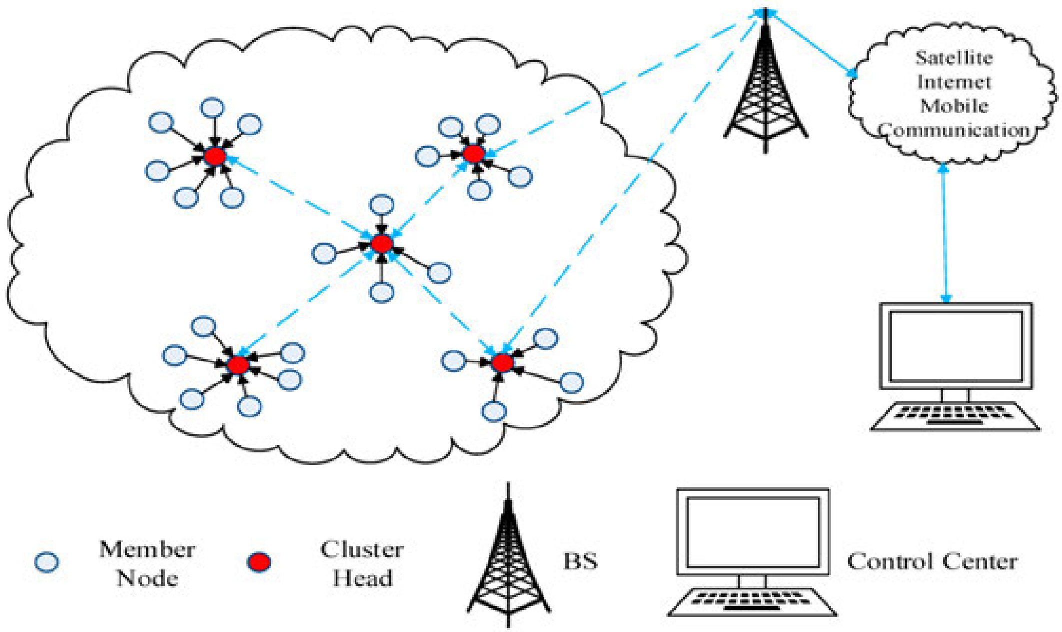


FIGURE 1. Structure of Industrial wireless sensor network.

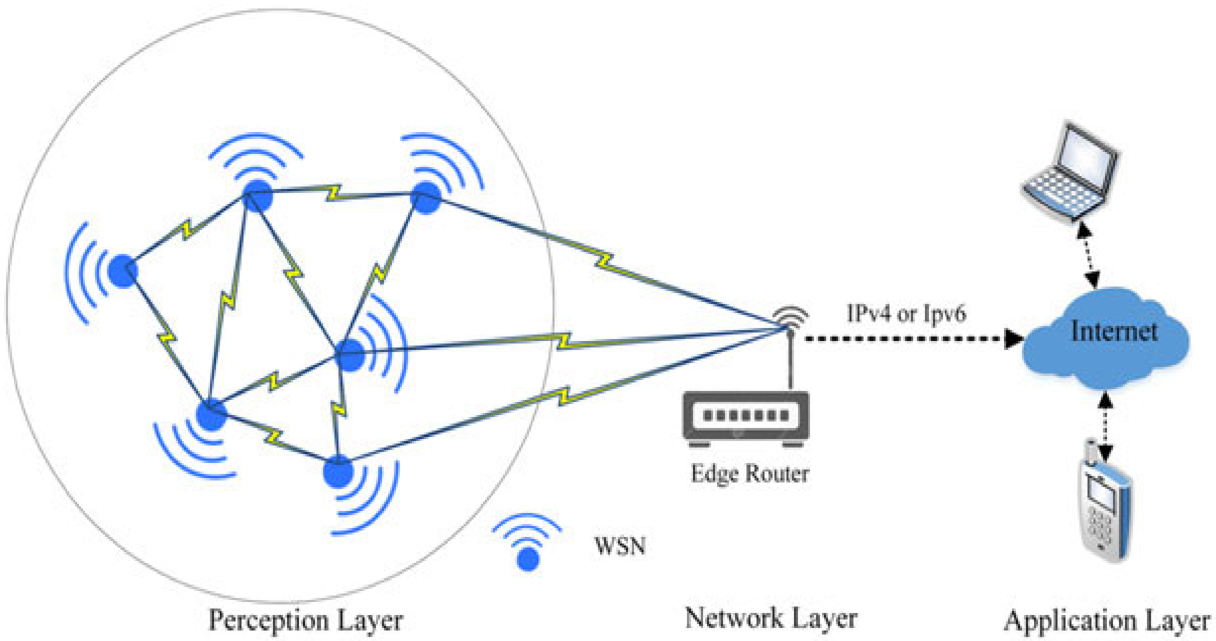


FIGURE 2. Layer configuration.

solar energy to power the system and send data over the network. This data is then read, collected, and stored by the machine learning algorithms embedded in the nodes. With the help of machine learning algorithms, communication efficiency can be improved by reducing the number of messages sent across sensor networks and improving the accuracy of the data sent. Furthermore, machine learning can be used to identify patterns in the data, allowing for quicker data classification and detection of security threats on the network.

Increasing Efficiency: The primary goal of an energy optimization model should be to maximize energy efficiency by reducing the overall energy costs and striving towards greater sustainability. Strategies such as LED lighting, energy-saving appliances, and alternative energy sources should all be considered.

Reducing Energy Wastage: An energy optimization model should ensure that any energy generated or consumed is used responsibly and efficiently. It includes assessing current usage patterns, identifying areas of potential waste or inefficiency, and putting in place measures such as timers, sensors, and remote monitoring to help ensure energy is being used efficiently.

Increasing User Comfort: An energy optimization model should strive to strike the right balance between comfort and cost saving. For example, reducing energy consumption by turning down thermostats might reduce costs but also decrease user comfort if the temperatures drop too far. Strategies such as insulation and investing in energy-efficient equipment should maximize user satisfaction while still achieving cost savings.

Resource Conservation: Resources are becoming increasingly scarce, so an energy optimization model should also strive to reduce resource consumption wherever possible. It could include improving efficiency through using renewable energy sources, recycling existing equipment and materials, and introducing energy-saving and innovative monitoring technologies.

B. PROPOSED ALGORITHM

The proposed model is based on an improved decentralized approach to communication. The nodes are organized in clusters, and the communication between nodes is done in a secure environment. The clusters in this edition of the enhanced energy optimization model for Industrial wireless sensor networks use machine learning algorithms that can identify abnormal behaviors in the communication process and detect potential threats. It provides greater security to the connections between the nodes, which can help protect the data transmitted across the network. The proposed model is a powerful technology that can provide more efficient, secure, and accurate communication between multiple nodes. The following algorithm.1 has shown the functions of the proposed model.

Algorithm 1 Enhanced energy optimization algorithm

1. Start
2. Compute the node 'i' and $F_{CH-Value}$
3. Generate random value between 0 and 1
4. If ($F_{CH-Value} > K$)
5. Then node 'i' can competes with other CHV nodes
6. the Node 'i' is non CHV
7. If the node 'i' is the winner
8. Then Node 'i' is CHV
9. Else go to step 6
10. End

This technology can also help improve the energy efficiency of wireless sensor networks, making them more cost-effective and efficient in the long run. The proposed model is an advanced model designed to reduce energy consumption in wireless sensor networks. The flow chart of the proposed model has shown in the following Fig.3

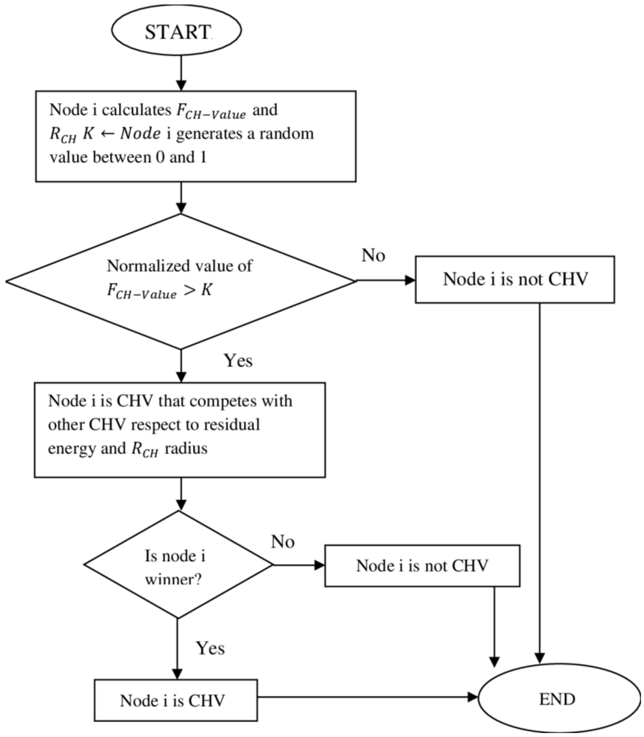


FIGURE 3. Flow diagram of an enhanced energy optimization model.

This model combines two cutting-edge technologies — machine learning and wireless sensors — to create a model that more effectively optimizes energy usage. The operating principle of the enhanced energy optimization model is based on predictive analytics. It uses machine learning algorithms to identify patterns in wireless sensors' energy usage and predict the amount of energy required for the desired operation. It ensures that the energy used is optimized for the specific task and can be minimized. Several algorithms and techniques are used to ensure that the model is both efficient and effective at reducing energy consumption. The model uses clustering and pattern recognition techniques to identify and predict the energy needs in areas with higher sensor concentrations. It allows the model to accurately predict the exact energy required for the operations of these clusters. Additionally, the model uses adaptive techniques to adjust the power requirements of the wireless sensors according to the changing conditions and demands of the external environment. It ensures that the energy usage is optimized even when the needs and conditions outside the sensor network change and allows for better energy optimization. Moreover, the model also uses reinforcement learning to learn from its mistakes and adapt to optimize energy usage during operation. It enables the model to adjust its energy settings based on the tasks and conditions, thus ensuring optimal energy consumption.

The proposed algorithm begins by splitting the data set into separate components. It can be done either randomly or through a predetermined grouping strategy. Then, for each component, the algorithm builds a model using Linear Regression and the Energy Measurement Technique, which minimizes the error of each model and maximizes the model accuracy for each component. The algorithm then move on to its optimization stage and tries to find the best combinations of models that minimize the overall error. It first selects two models from each component that minimize the error and maximizes the model accuracy. It then creates the first combination by merging those two models from different components and calculating the resulting total error and model accuracy. It then moves on to the second combination, selecting the following two models from each component and creating the new combination. This process is repeated until the minimum error value is found. Once the algorithm finds the best combination, it applies the Energy Measurement Technique to this combination to improve the total model accuracy and minimize the total error. It is done by optimizing the parameter values in the combination to maximize the model's accuracy. Finally, the algorithm outputs the optimized combination of models and their associated parameter values.

The proposed model is based on predictive analytics and can be used to predict and optimize energy consumption in wireless sensor networks accurately. The model can effectively reduce energy consumption while ensuring efficient operation by using various algorithms and techniques, such as clustering, pattern recognition, adaptive techniques, and reinforcement learning. This research aims to construct an enhanced energy optimization model for industrial wireless sensor networks using machine learning. In order to achieve this, the research will include both experimental and theoretical approaches to identify significant parameters and develop the energy optimization model. This model will consider energy consumed by wireless communication and the sensing and data processing costs, as well as the trade-offs between the amount of energy consumed and the accuracy of the sensed data. First, experiments will be conducted in an industrial laboratory environment by connecting various types of wireless sensors to an industrial system.

C. SIGNIFICANCE OF THE MODEL

The Enhanced Energy Optimization Model (EEOM) is an innovative approach that utilizes machine learning and artificial intelligence to optimize energy in industrial wireless sensor networks. It is based on a holistic approach, combining techniques from signal processing, control theory, energy management, and machine learning, intending to provide high energy efficiency and power savings. WSNs consist of multiple sensors, which communicate wirelessly in a distributed and self-organized manner. It takes in input data collected from the sensors in the WSN, which can be used to build energy usage profiles and predict future energy consumption scenarios. This model uses sensor data to create accurate energy usage predictions, and these predictions can be used to plan a more efficient energy usage strategy or optimize the existing one. The primary technical significance of EEOM is that it can monitor energy usage patterns and adjust energy levels accordingly to maximize energy efficiency. The proposed model can incorporate different data collected from the WSNs, such as temperature and humidity levels. It assists in achieving optimal energy performance and reduces the need to monitor energy levels manually. It addresses the energy issues of such networks by using advanced data processing and modeling techniques to predict energy consumption and optimize the operation of the network. Using machine learning, this model can more accurately identify energy-saving opportunities and reduce the loss due to inefficiencies. This Model's ability to better analyze data and identify patterns can help improve system performance, such as faster data transmission and more reliable connections. Moreover, it can predict future energy usage trends and detect anomalies or deviations from the typical use pattern. It helps avoid energy wastage and makes the industrial environment more energy-efficient.

D. SENSITIVITY ANALYSIS

Sensitivity analysis is a technique used to identify how the outputs of an energy optimization model may change when different factors are varied. The technique is used to determine which factors can be modified in order to enhance energy optimization. This can include looking at the degree of improvement for changing different parameters within the model, such as the types of building materials used, the efficiency of the heating and cooling system, and changes to the insulation of a building. Through this analysis, improvements can be identified which can reduce the total energy consumption associated with the building, while still maintaining a comfortable environment inside. The changes to set parameters, input variables, or existing functions used in an energy optimization model can significantly affect its performance. If the parameters, input variables, or existing functions are altered, they may reduce the accuracy of the energy optimization model or make it harder to achieve an optimal solution. Consequently, it is essential to thoroughly assess the impacts of any changes to ensure that the model functions as expected. The assessment should include testing the energy optimization model with different combinations of parameters, input variables, and existing functions to evaluate the performance and determine the best approach for achieving an optimal result. The model relies on these values to accurately simulate and predict energy use in the future. For example, if specific parameters are set incorrectly or the input variables are not appropriate for the studied area, the model's predictions may be accurate and accurate. Similarly, if existing functions are modified, the model may not effectively capture the energy use of specific devices or regions, leading to a less effective energy optimization plan. Thus, It is essential to ensure that the parameters, input variables, and existing functions are accurate and up-to-date for the model to predict and optimize energy use accurately.

IV. ANALYTICAL DISCUSSIONS

The proposed model is a powerful application of advanced machine learning techniques for efficiently powering the sensors in industrial settings. This advanced model leveraging machine learning methods offers a novel power optimization approach to optimize energy consumption and ensure a continuous flow of data from the sensor nodes in an industrial network. The sensors can be managed autonomously to minimize energy usage and meet the application's system requirements. This model also enables improved performance by reducing the power requirements of all the sensors on an industrial network. The model works by optimizing the transmission settings of all the sensors in a network or cluster of networks. It implements a variety of techniques in order to configure all sensors while also considering their energy characteristics correctly. The model then uses sophisticated algorithms to determine the optimal transmission parameters for each sensor. Let us see if the communication node has the received energy (r) and the transmitted energy (t). Then it has the channel matrix 'm' with the additive Gaussian noise (n) listed below eq.1

$$r = (m * t) + n \tag{1}$$

Now the computation of output of first sensor node in the network in initial cluster has the following eq.2

$$S_a^1 = t_a * \left(\sum_{p=1}^{n} W_{1p} * t_p + \beta_p \right) \tag{2}$$

where, W_{1p} denotes the weight of the cluster, β_p indicates the threshold co-efficient of the sensor network. t_a represents the activation function. Now the output of the final sensor has the following eq.3

$$S_a^f = t_a^f * \left(W^f * t^f + \beta^f \right) \tag{3}$$

The model also uses machine learning techniques to optimize the settings of each sensor in the network, ensuring that the energy consumption is minimized while still meeting the needs of the network and application. The enhanced energy optimization model considers the changes in the environment and state of the sensor nodes, allowing it to adapt quickly and adjust its features accordingly. The performance of the enhanced energy optimization model can be further improved by studying the machine learning algorithms used in the model. By understanding the limitations of these methods, one can ensure that the model produces an optimal output. Additionally, the model can be tuned for specific environments, such as in an industrial setting. It will help ensure that the model can adjust its settings to meet the application's specific needs, which may require adjusting the frequency of transmission or the data rates. The proposed model offers a robust solution for optimizing energy usage and ensuring a continuous data flow in industrial settings. By leveraging advanced machine learning algorithms and optimizing transmission settings, the model can produce optimal output and be used to maximize the performance of a network or cluster of networks.

V. COMPARATIVE ANALYSIS

The performance of the proposed enhanced energy optimization model (EEOM) has compared with the existing Deep-learning-based physical layer (DLBPL), Deep Learning-Assisted Smart Process Planning (DLASP), Energy-efficient algorithm (EEA) and Outlier detection in wireless sensor networks (ODWSN). Here the network simulator (NS-2) is the tool used to execute the results. Here the industrial wireless sensor networks dataset [39] has been taken for comparison. Totally of 7846 samples have been taken for the analysis. In that, 70% (5492 samples) were used for training, and 30% (2354 samples) were used for testing. Table.2 shows the simulation parameters.

TABLE 2. Simulation parameter.

PARAMETER	VALUE
Estimated simulation Area	1150m×1150m
SIFS (Short Inter-Frame Space)	35 sec
DR_T (Data Rate for Transmission)	240 kbps
IDR (Interference detection rate)	70 ms
T_P (Primary slot time)	15 ms
B_S (Bandwidth for Sub-channel)	220 MHz
B_C (Bandwidth for Channel)	120 MHz
F_C (Carrier frequency)	32.46 MHz
T_S (Simulation duration)	2 min (120 sec)

A. COMPUTATION OF TRANSMIT ENERGY CONSUMPTION (TEC)

The potential for machine learning to reduce transmit energy consumption in industrial wireless sensor networks (IWSNs) is substantial. Firstly, machine learning can be used to identify patterns in the data collected from the sensors. By analyzing the data, machine learning algorithms can detect anomalies and inefficiencies in the system and suggest changes that can reduce energy consumption. The proposed algorithm can identify when a sensor is sending redundant or unnecessary data or identify when a sensor can be powered down until the data is needed.

$$TEC = \{C_t * V_t * T_t\} \tag{4}$$

where C_t expresses the transmit current, V_t indicates the applied voltage at the transmitter end, and T_t denotes the transmit time between the sensor nodes. The machine learning algorithms can be used to optimize the transmission power of the nodes in the IWSN. By using an algorithm to adjust the transmit power of each node, the network can reduce its energy consumption while still ensuring that data is successfully transmitted.

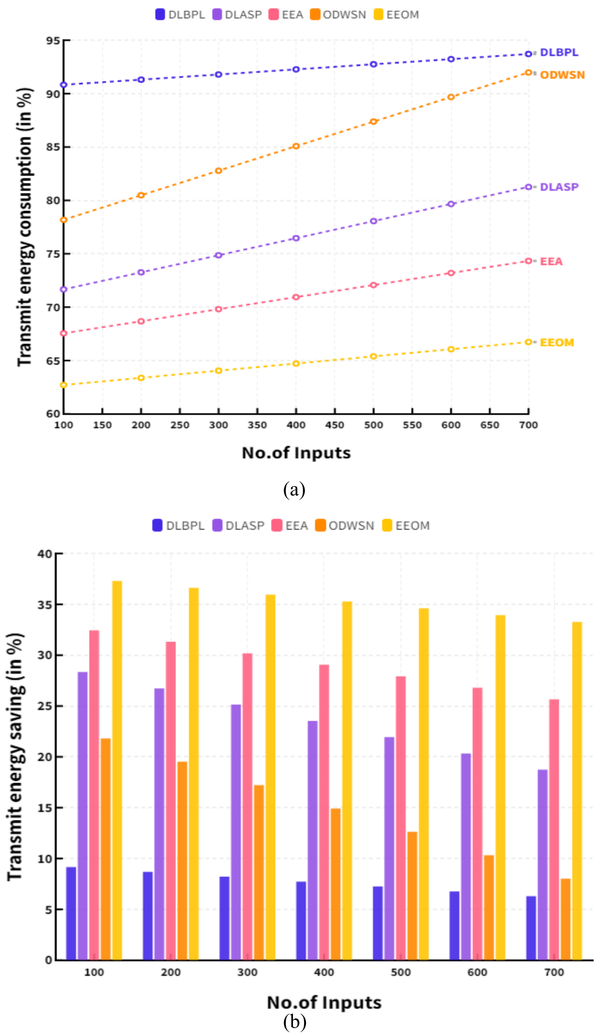


FIGURE 4. (a) Estimation of transmit energy consumption. (b) Estimation of transmit energy saving.

Fig.4 shows the estimation of transmit energy between existing and proposed models. Fig. 4. (a) Indicates the transmit energy consumption, and Fig. 4. (b) Represents the transmit energy saving. The proposed algorithm is designed to reduce the energy consumption of the sensor nodes by optimizing the transmission power. The proposed can be applied to the scheduling of transmissions in the IWSNs. By predicting the optimal times for each node to transmit, the network can minimize its energy consumption by scheduling transmissions when the conditions are most favorable. Table.3 shows the estimation of transmit energy consumption between the existing and proposed models. The model relies on machine learning algorithms to identify the most efficient transmission power for each node based on the data it collects from the environment. The machine learning algorithm can learn patterns in the data and adjust the transmission power accordingly, reducing the energy required. This model can help reduce the network's energy consumption by up to 40%.

B. COMPUTATION OF RECEIVED ENERGY CONSUMPTION (REC)

Machine learning can be used to optimize energy consumption in industrial wireless sensor networks by providing predictive analytics that can accurately predict the energy needs of the network. The proposed algorithm can be trained to identify patterns in energy use and make predictions about future needs. By understanding the energy usage of the network, the algorithm can then recommend changes to the network's energy management system to reduce the overall energy consumption.

$$REC = \{C_r * V_r * T_r\} \tag{5}$$

where, C_r express the received current, V_r indicates the applied voltage at the receiver end and T_r denotes the receiver time between the sensor nodes. Table.4 shows the estimation of received energy consumption between the existing and proposed models.

Fig.5 shows the estimation of received energy between existing and proposed models.

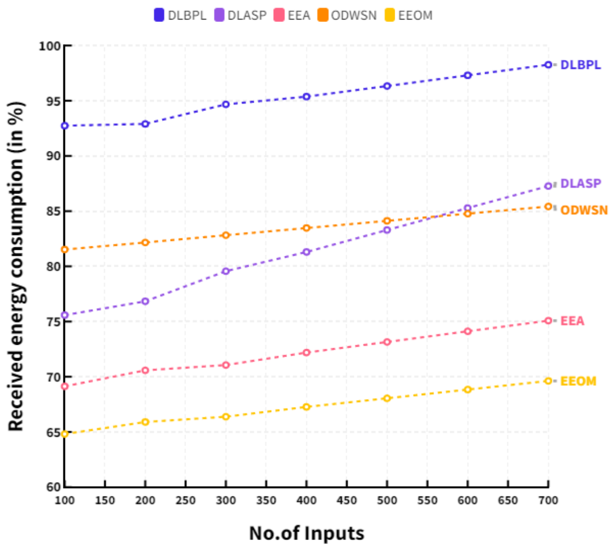
Where Fig. 5.(a) indicated the received energy consumption and fig.5.(b) represents the received energy saving. The machine learning can be used to identify and optimize inefficient energy usage, such as discovering when sensors are not sending out data or when transmission rates are too high. This can help reduce the overall energy consumption of the network while still providing accurate data.

C. COMPUTATION OF IDLE MODE ENERGY CONSUMPTION (IEC)

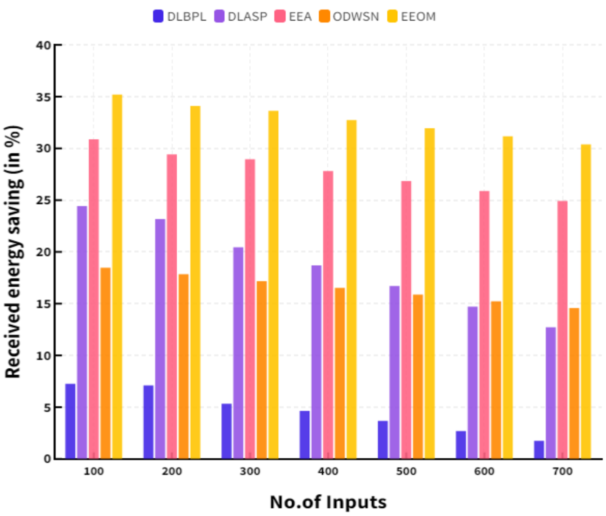
Using machine learning, industrial wireless sensor networks can accurately predict the energy consumption of different idle states and take the necessary steps to reduce it. This can be done by analyzing the data collected from the sensor nodes, looking for patterns in the data that can be used to determine how much energy are being consumed in different idle states. This can then be used to set up rules for automatically adjusting the power consumption of the nodes to the optimal level for the given idle state. This leads to a reduction in energy consumption and an increase in overall efficiency.

$$IEC = \{C_i * V_i * T_i\} \tag{6}$$

where, Ci express the idle mode current, Vi indicates the applied voltage at the idle mode and Ti denotes the idle time between the sensor nodes. Table.5 shows the estimation of idle-mode energy consumption between the existing and proposed models. Fig.6 shows the estimation of idle-mode energy between existing and proposed models. Where Fig. 6.(a) indicated the idle-mode energy consumption and fig.6.(b) represents the idle-mode energy saving. Different data rates and energy intensities of the wireless communication will be tested on the industrial system to

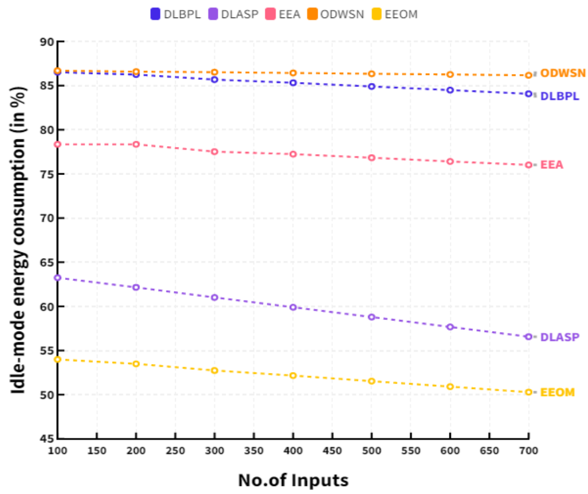


(a)

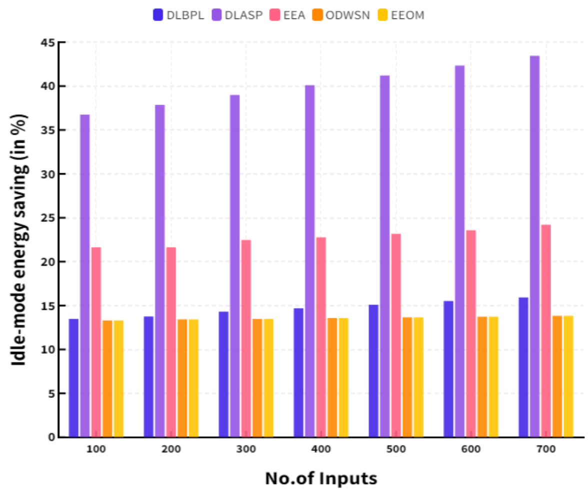


(b)

FIGURE 5. **(a) Stimation of received energy consumption. (b) Estimation of received energy saving.**



(a)



(b)

FIGURE 6. **(a) Estimation of idle-mode energy consumption. (b) Estimation of idle-mode energy saving.**

investigate how varying energy-intensity affects the total energy consumption. The experimental results will be used to calibrate the parameters that best describe the energy-information trade-off of the industrial system.

D. COMPUTATION OF SLEEP MODE ENERGY CONSUMPTION (SEC)

The use of machine learning to optimize the sleep state energy consumption of industrial wireless sensor networks has become increasingly popular in recent years. Machine learning algorithms can be trained to identify when it is most appropriate to enter and exit the sleep state, while minimizing energy consumption and maximizing performance. By leveraging the power of machine learning, it is possible to tailor the sleep state energy consumption of industrial wireless sensor networks to specific user and application needs. With machine learning, the sleep state energy consumption can be optimized by taking into account factors such as the time of day, the location of the sensor, the type of application, and the network traffic. In addition, machine learning can be used to detect anomalies and provide predictive maintenance solutions to extend the lifetime of industrial wireless sensor networks.

$$SEC = \{C_s * V_s * T_s\} \tag{7}$$

where, Cs express the sleep mode current, Vs indicates the applied voltage at the sleep mode and Ts denotes the sleep time between the sensor nodes. Table.6 shows the estimation of sleep-mode energy consumption between the existing and proposed models.

Fig.7 shows the estimation of sleep-mode energy between existing and proposed models. Where Fig. 7.(a) indicated the sleep-mode energy consumption and fig.7.(b) represents the sleep-mode energy saving. In the theoretical part of the

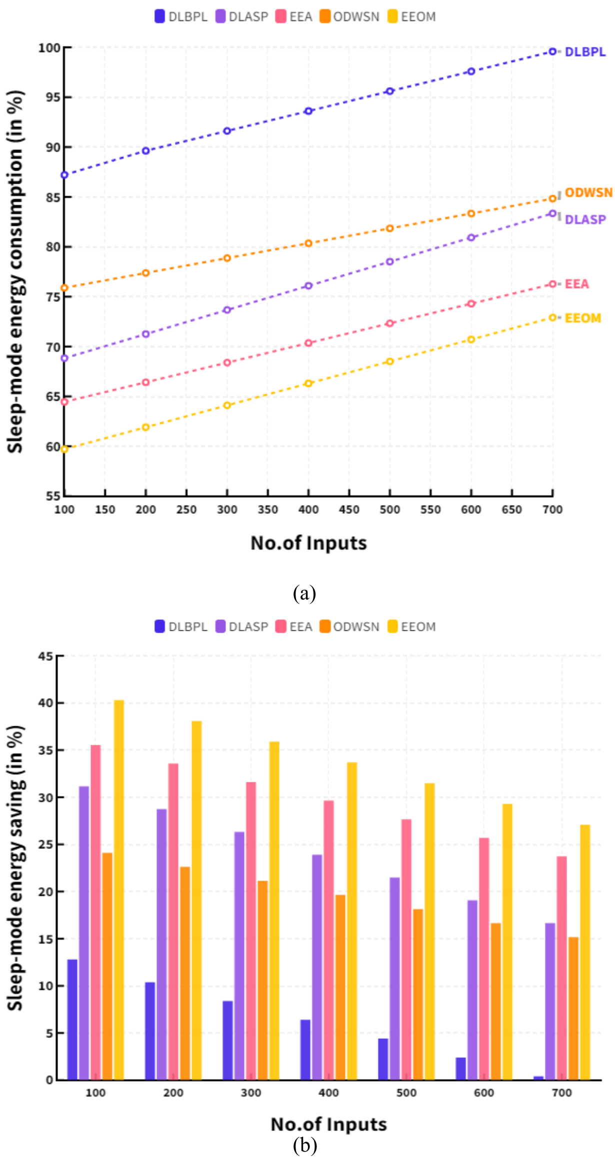


FIGURE 7. **(a) Estimation of sleep-mode energy saving. (b) Estimation of sleep-mode energy consumption.**

project, a computational model will be used to define the energy optimization problem, in which different costs associated with the wireless communication and data processing will be taken into account. At the same time, machine learning techniques will be utilized to come up with optimized solutions for the energy optimization problem by taking into account actual sensor data gathered in the experiments. This approach will also include methods for distinguishing between noise and meaningful data collected from the sensors.

E. COMPUTATION OF PREVALENCE THRESHOLD (P_{th})

The computation of a prevalence threshold for an enhanced energy optimization model for industrial wireless sensor networks using machine learning involves using an AI-assisted approach to identify an appropriate risk threshold for an industrial wireless sensor network. This approach builds upon existing energy optimization models using machine learning techniques to leverage the existing energy optimization model for the industrial environment. A risk threshold can be identified by incorporating the knowledge of the current industrial environment and other machine-learning techniques, such as natural language processing. In other words, a prevalence threshold can be determined by identifying the optimal risk score for a particular industrial environment using machine learning approach such as a decision tree or grey scale clustering. The prevalence threshold has computed with the help of following eq.8

$$P_{th} = \left\{ \frac{\sqrt{PR_{false}}}{\sqrt{PR_{true}} + \sqrt{PR_{false}}} \right\} \tag{8}$$

where, P_{th} indicates the prevalence threshold, PR_{false} represents the positive rates in false predictions and PR_{true} represents the positive rates in true predictions. Table.7 shows the comparison of prevalence threshold between the existing and proposed models.

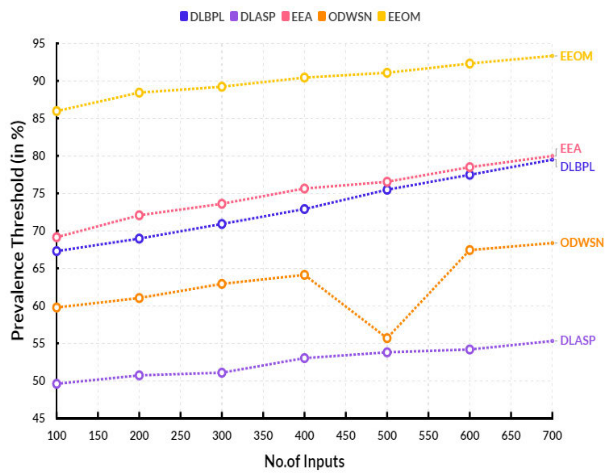


FIGURE 8. **Estimation of prevalence threshold.**

The selected risk threshold can then be applied to the energy optimization model to identify the best possible energy optimization for a given industrial environment. Fig.8 shows the comparison of prevalence threshold. In a comparison point the proposed EEOM obtained 90.44% prevalence threshold. The existing DLBPL reached 72.90%, DLASP secured 53.02%, EEA obtained 75.64% and ODWSN reached 64.12% prevalence threshold. The sudden drop in ODWSN caused by interference from other wireless devices in the vicinity. For example, a nearby Wi-Fi or cellular connection could generate radio frequency interference that could make it difficult for nodes within the sensor network to communicate properly. Additionally, if the sensor nodes are powered by batteries, their lifespan could be affected and cause the outlier detection system to experience a drop in performance. Finally, faulty or poorly maintained wiring can also cause data readouts to become unreliable and affect the outlier detection system.

F. COMPUTATION OF CRITICAL SUCCESS INDEX (CSI)

The Critical Success Index (CSI) measures the proposed model in achieving its objectives. The CSI assesses the model's effectiveness and efficiency in utilizing the available resources to achieve the desired goal or outcome within a set timeline. The process of calculating the CSI involves the following steps:

- Identify and define the criteria that will be used to assess the success of the proposed model. It could include energy efficiency, network coverage, and response time.
- Measure the actual performance of the proposed using the identified criteria.
- Calculate the weighted average of the criteria using a weighting system.
- Calculate the Critical Success Index (CSI) by subtracting the weighted average from the theoretical perfect score 100.

$$CSI = \left\{ \frac{TP}{TP + FN + FP} \right\} \tag{9}$$

Table.8 shows the comparison of critical success index between the existing and proposed models. Fig.9 shows the comparison of critical success index. In a comparison point the proposed EEOM obtained 90.33% critical success index. The existing DLBPL reached 72.28%, DLASP secured 48.41%, EEA obtained 75.80% and ODWSN reached 63.80% critical success index. The calculation of the CSI will indicate how well the Enhanced Energy Optimization Model for Industrial Wireless Sensor Networks using Machine Learning is performing and should be used as an indicator of the progress made towards its set objectives.

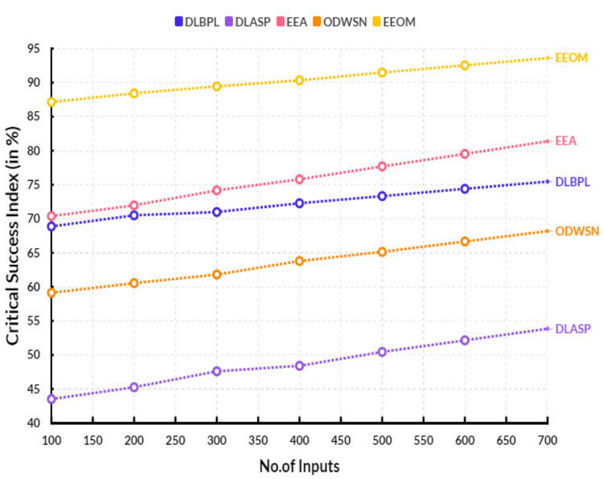


FIGURE 9. Estimation of Critical Success Index.

G. COMPUTATION OF DELTA-P (ΔP)

Delta-P is the term used to describe an industrial Wireless Sensor Network (WSN) performance optimization. It measures the difference between the actual power consumption of a WSN and its optimal power consumption. Delta-P is typically calculated by multiplying the difference between the optimum and actual energy by 100 and dividing by the optimum energy. Delta-P has computed with the help of eq.10

$$\Delta P = PPV - NPV - 1 \tag{10}$$

where, PPV = positive predicted values, NPV = negative predicted values. Table.9 shows the comparison of Delta-P between the existing and proposed models.

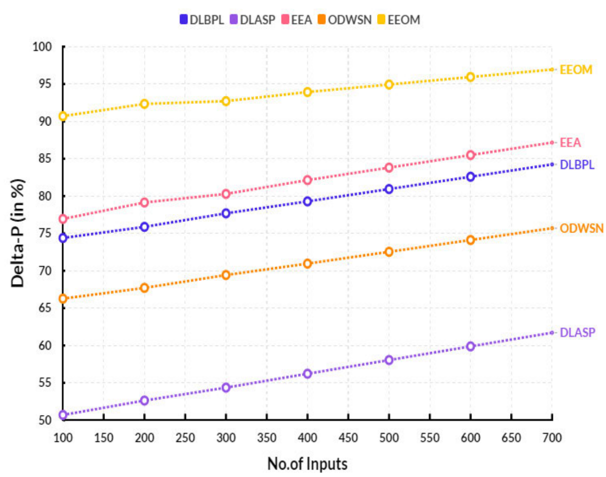


FIGURE 10. Estimation of Delta-P.

Fig.10 shows the comparison of Delta-P. In a comparison point the proposed EEOM obtained 93.93% Delta-P. The existing DLBPL reached 79.29%, DLASP secured 56.22%, EEA obtained 82.14% and ODWSN reached 70.96% Delta-P. The calculation is used to determine the relative efficiency of the power system and can be used to identify areas of improvement. Delta-P is often employed compared to a base-case energy optimization solution, such as a single WSN node model. It allows for comparing different configurations, intending to optimize energy efficiency.

H. COMPUTATION OF MATTHEWS'S CORRELATION COEFFICIENT (MCC)

Matthews' correlation coefficient measures the correlation between the observed and predicted values of a given data set. It measures the quality of the predictions that have been made. Matthews' correlation coefficient is also called the coefficient of determination (R^2). In the context of an enhanced energy optimization model for industrial wireless sensor networks, the Matthews' correlation coefficient is used to measure the accuracy of the prediction made by the model. It is beneficial when several variables are being assessed. The Matthews' correlation coefficient compares the data's actual observed values to the model's predicted values. It considers the prediction's false positives, false negatives, and true positives and negatives. The formula to calculate the Matthews' correlation coefficient is,

$$MCC = \left\{ \frac{(TP * TN) - (FP * FN)}{\sqrt{(TP + FP) * (TP + FN) * (TN + FP) * (TN + FN)}} \right\} \quad (11)$$

where: TP = True positives, TN = True negatives, FP = False positives, and FN = False negatives. Table.10 shows the comparison of MCC between the existing and proposed models.

Fig.11 shows the comparison of MCC. In a comparison point the proposed EEOM obtained 90.06% MCC. The existing DLBPL reached 62.54%, DLASP secured 52.41%, EEA obtained 75.13% and ODWSN reached 64.87% MCC. The Matthews' correlation coefficient can be used to evaluate the performance of the energy optimization model in industrial wireless sensor networks. Its value indicates the accuracy of the model in predicting the correct outcome. A higher Matthews' correlation coefficient indicates a better prediction accuracy of the model.

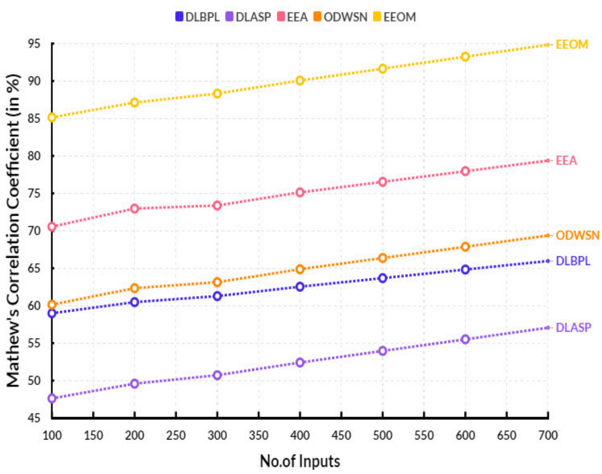


FIGURE 11. Estimation of Matthews' correlation coefficient.

I. COMPUTATION OF FOWLKES–MALLOWS INDEX (FMI)

The Fowlkes–Mallows index measures the similarity between two different clustering results. It is calculated by measuring the similarity between two sets of clusters, A and B, by using the formula:

$$FMI = \sqrt{\left(\frac{TP}{TP + FP}\right) * \left(\frac{TP}{TP + FN}\right)} \quad (12)$$

where, TP is the number of true positives (data points that were correctly identified as belonging to the same group in both sets) and FP and FN are the false positives and false negatives. Table.11 shows the comparison of FMI between the existing and proposed models.

Fig.12 shows the comparison of FMI. In a comparison point the proposed EEOM obtained 92.17% FMI. The existing DLBPL reached 65.92%, DLASP secured 54.33%, EEA obtained 81.28% and ODWSN reached 66.81% FMI. The higher the FM index, the closer the clustering results are to each other. For the Enhanced Energy Optimization Model for Industrial Wireless Sensor Networks, the Fowlkes–Mallows index could be used to measure how well the energy optimization model improves the network's performance in terms of energy optimization and clustering results. The higher the Fowlkes–Mallows index, the more accurate the model.

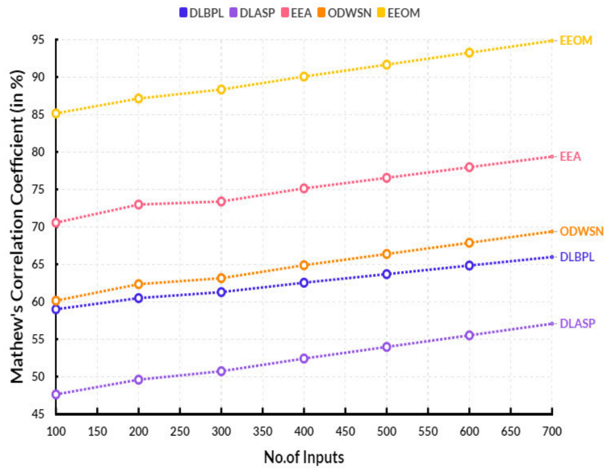


FIGURE 12. Estimation of Fowlkes–Mallows index.

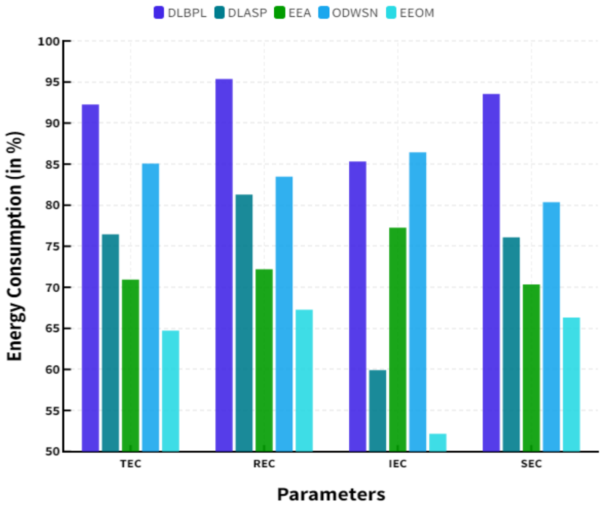
VI. RESULTS AND DISCUSSION

The research efforts will culminate in formulating an efficient energy optimization model for industrial wireless sensor networks based on the results of the experiments and the theoretical work. The model will serve as a guide to the design and operation of existing and future industrial wireless systems.

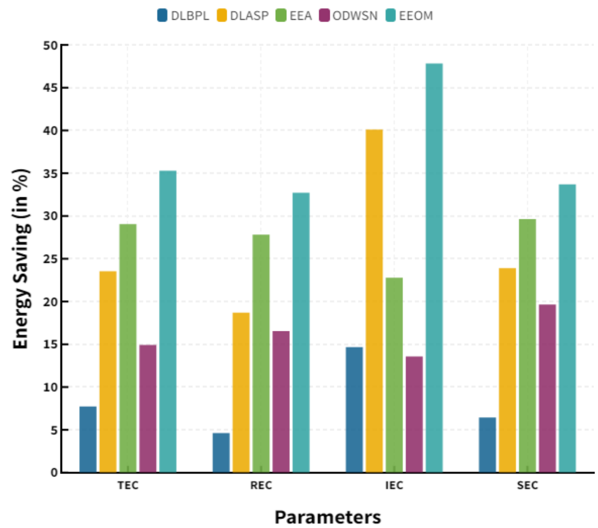
Table.12 shows the overall performance comparison. Table.13 shows the overall energy consumption and energy saving. It is expected that applying the proposed energy optimization model will improve the energy efficiency of industrial wireless networks, contributing to a more sustainable energy industry. Fig.13 shows the estimation of overall performance between existing and proposed models. Fig. 13. (a) Indicates the energy consumption, and Fig. 13. (b) Represents the energy saving. Fig.13.(c) indicates the overall performance comparison. In a comparison point, the proposed EEOM obtained 64.72% transmission energy consumption, 35.28% transmission energy saving, 67.27% received energy consumption, 32.73% received energy storage, 52.16% idle-mode energy consumption, 47.84% idle-mode energy storage, 66.31% sleep-mode energy consumption, and 33.69% sleep-mode energy storage.

In a comparison point, the proposed EEOM obtained 90.44% prevalence threshold, 90.33% critical success index, 93.93% Delta-P, 90.06% MCC and 92.17% FMI rates.

The existing DLBPL obtained 92.27% transmission energy consumption, 7.73% transmission energy saving,



(a)



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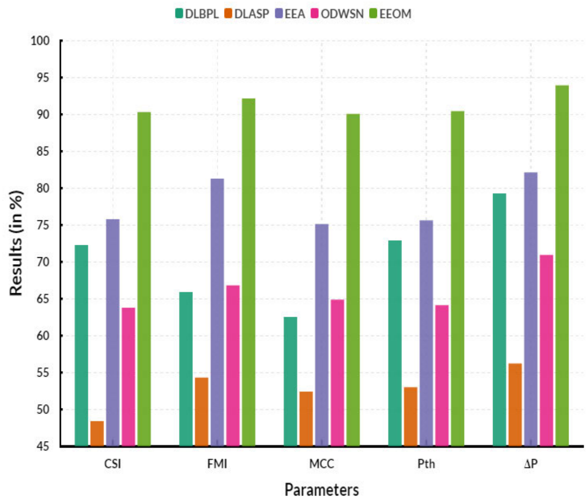

(c)

FIGURE 13. (a) Estimation of overall energy consumption. (b) Estimation of overall energy saving. (c) Estimation of overall performance.

TABLE 3. Estimation of transmit energy consumption (in %).

No.of Inputs	Allotted Energy (in %)	DLBPL		DLASP		EEA		ODWSN		EEOM	
		(C)	(S)	(C)	(S)	(C)	(S)	(C)	(S)	(C)	(S)
100	100	90.83	9.17	71.66	28.34	67.55	32.45	78.18	21.82	62.71	37.29
200	100	91.31	8.69	73.26	26.74	68.68	31.32	80.48	19.52	63.38	36.62
300	100	91.79	8.21	74.86	25.14	69.81	30.19	82.78	17.22	64.05	35.95
400	100	92.27	7.73	76.46	23.54	70.94	29.06	85.08	14.92	64.72	35.28
500	100	92.75	7.25	78.06	21.94	72.07	27.93	87.38	12.62	65.39	34.61
600	100	93.23	6.77	79.66	20.34	73.2	26.8	89.68	10.32	66.06	33.94
700	100	93.71	6.29	81.26	18.74	74.33	25.67	91.98	8.02	66.73	33.27

95.37% received energy consumption, 4.63% received energy storage, 85.32% idle-mode energy consumption, 14.68% idle-mode energy storage, 93.55% sleep-mode energy consumption, and 6.45% sleep-mode energy storage. The existing DLASP obtained 76.46% transmission energy consumption, 23.54% transmission energy saving, 81.30% received energy consumption, 18.70% received energy storage, 59.90% idle-mode energy consumption,

TABLE 4. Estimation of received energy consumption (in %).

No.of Inputs	Allotted Energy (in %)	DLBPL		DLASP		EEA		ODWSN		EEOM	
		(C)	(S)	(C)	(S)	(C)	(S)	(C)	(S)	(C)	(S)
100	100	92.74	7.26	75.58	24.42	69.13	30.87	81.52	18.48	64.81	35.19
200	100	92.9	7.1	76.83	23.17	70.58	29.42	82.16	17.84	65.9	34.1
300	100	94.67	5.33	79.56	20.44	71.06	28.94	82.82	17.18	66.38	33.62
400	100	95.37	4.63	81.3	18.7	72.19	27.81	83.47	16.53	67.27	32.73
500	100	96.33	3.67	83.29	16.71	73.15	26.85	84.12	15.88	68.05	31.95
600	100	97.3	2.7	85.28	14.72	74.12	25.88	84.77	15.23	68.84	31.16
700	100	98.26	1.74	87.27	12.73	75.08	24.92	85.42	14.58	69.62	30.38

TABLE 5. Estimation of Idle-mode energy consumption (in %).

No.of Inputs	Allotted Energy (in %)	DLBPL		DLASP		EEA		ODWSN		EEOM	
		(C)	(S)	(C)	(S)	(C)	(S)	(C)	(S)	(C)	(S)
100	100	86.51	13.49	63.24	36.76	78.35	21.65	86.69	13.31	53.99	46.01
200	100	86.25	13.75	62.15	37.85	78.35	21.65	86.58	13.42	53.49	46.51
300	100	85.68	14.32	61.01	38.99	77.52	22.48	86.52	13.48	52.74	47.26
400	100	85.32	14.68	59.9	40.1	77.24	22.76	86.43	13.57	52.16	47.84
500	100	84.9	15.1	58.79	41.21	76.83	23.17	86.34	13.66	51.53	48.47
600	100	84.49	15.51	57.67	42.33	76.41	23.59	86.26	13.74	50.91	49.09
700	100	84.07	15.93	56.56	43.44	76.02	24.21	86.17	13.83	50.28	49.72

TABLE 6. Estimation of sleep-mode energy consumption (in %).

No.of Inputs	Allotted Energy (in %)	DLBPL		DLASP		EEA		ODWSN		EEOM	
		(C)	(S)	(C)	(S)	(C)	(S)	(C)	(S)	(C)	(S)
100	100	87.21	12.79	68.83	31.17	64.45	35.55	75.89	24.11	59.71	40.29
200	100	89.63	10.37	71.25	28.75	66.42	33.58	77.38	22.62	61.91	38.09
300	100	91.62	8.38	73.67	26.33	68.39	31.61	78.87	21.13	64.11	35.89
400	100	93.61	6.39	76.09	23.91	70.36	29.64	80.36	19.64	66.31	33.69
500	100	95.6	4.4	78.51	21.49	72.33	27.67	81.85	18.15	68.51	31.49
600	100	97.59	2.41	80.93	19.07	74.3	25.7	83.34	16.66	70.71	29.29
700	100	99.58	0.42	83.35	16.65	76.27	23.73	84.83	15.17	72.91	27.09

40.10% idle-mode energy storage, 76.09% sleep-mode energy consumption, and 23.91% sleep-mode energy storage. The existing EEA obtained 70.94% transmission energy consumption, 29.06% transmission energy saving, 72.19% received energy consumption, 27.81% received energy storage, 77.25% idle-mode energy consumption, 22.79% idle-mode energy storage, 70.36% sleep-mode energy consumption, and 29.64% sleep-mode energy storage. The existing ODWSN obtained 85.08% transmission energy consumption, 14.92% transmission energy saving, 83.47% received energy consumption, 16.53% received energy storage, 86.43% idle-mode energy consumption, 13.57% idle-mode energy storage, 80.36% sleep-mode energy consumption, and 19.64% sleep-mode energy storage.

Using machine learning, industrial wireless sensor networks (IWSNs) can have many applications. Machine learning can identify problems and anomalies in IWSNs, such as changes in temperature, pressure, vibration, and other physical parameters. It can also be used to analyze and predict the future behavior of sensors and devices and detect and classify different kinds of events. Machine learning can also enable IWSNs to self-configure, self-heal, and self-optimize to provide timely, accurate, and reliable data. Moreover, machine learning can be used to enable IWSNs to detect and respond to cyber-attacks. The effectiveness of the proposed model in industry practice will ultimately depend on the specific application and requirements. The model has the potential to provide several benefits, such as improved battery life, flexibility, scalability, and enhanced security. Depending on how carefully these benefits are taken advantage of, this model could lead to better energy efficiency in industrial monitoring and control systems, higher accuracy in data gathering, and cost savings on power consumption and system maintenance. The model may reduce energy consumption and improve the scalability of wireless sensor networks. However, it is up to the individual industrial entities to assess its effectiveness and determine the necessary level of adaptation.

TABLE 7. Estimation of prevalence threshold (in %).

No.of Inputs	DLBPL	DLASP	EEA	ODWSN	EEOM
100	67.29	49.59	69.14	59.77	85.97
200	68.96	50.72	72.07	61.03	88.43
300	70.91	51.07	73.60	62.92	89.22
400	72.90	53.02	75.64	64.12	90.44
500	75.48	53.79	76.54	55.68	91.08
600	77.47	54.17	78.49	67.43	92.31
700	79.49	55.30	79.98	68.36	93.33

TABLE 8. Estimation of critical success index (in %).

No.of Inputs	DLBPL	DLASP	EEA	ODWSN	EEOM
100	68.88	43.52	70.39	59.13	87.14
200	70.51	45.26	71.97	60.55	88.43
300	70.99	47.60	74.17	61.81	89.44
400	72.28	48.41	75.80	63.80	90.33
500	73.33	50.45	77.69	65.14	91.48
600	74.40	52.15	79.53	66.67	92.54
700	75.47	53.85	81.38	68.19	93.60

TABLE 9. Estimation of delta-P (in %).

No.of Inputs	DLBPL	DLASP	EEA	ODWSN	EEOM
100	74.39	50.70	76.94	66.27	90.70
200	75.88	52.63	79.14	67.71	92.34
300	77.69	54.36	80.29	69.43	92.71
400	79.29	56.22	82.14	70.96	93.93
500	80.94	58.05	83.81	72.54	94.93
600	82.59	59.88	85.49	74.12	95.94
700	84.24	61.71	87.16	75.70	96.94

TABLE 10. Estimation of matthews's correlation coefficient (in %).

No.of Inputs	DLBPL	DLASP	EEA	ODWSN	EEOM
100	58.99	47.62	70.55	60.14	85.14
200	60.48	49.59	72.97	62.34	87.13
300	61.28	50.72	73.38	63.14	88.33
400	62.54	52.41	75.13	64.87	90.06
500	63.68	53.96	76.54	66.37	91.65
600	64.83	55.51	77.96	67.87	93.25
700	65.97	57.06	79.37	69.37	94.84

The technical limitations of an Enhanced Energy Optimization Model for Industrial Wireless Sensor Networks include limited energy, limited bandwidth, limited signal strength, and difficulty in managing data transmission. Likewise, scalability is also an issue. In addition, due to the wireless nature of the network, the transmission of energy-intensive data can lead to network congestion, preventing the messages from being received promptly. Furthermore, insecure protocols may introduce security vulnerabilities due to a lack of authentication and encryption. Finally, if the network is simple enough, it can introduce latency and fragmentation issues, ultimately reducing communication performance. The proposed enhanced energy optimization mode work faces some challenges. They are,

TABLE 11. Estimation of Fowlkes–Mallows index (in %).

No.of Inputs	DLBPL	DLASP	EEA	ODWSN	EEOM
100	63.61	51.93	74.98	63.81	88.81
200	64.62	52.30	77.30	65.24	90.24
300	65.26	53.83	78.55	66.33	91.40
400	65.92	54.33	81.28	66.81	92.17
500	66.74	55.28	83.06	68.07	93.46
600	67.50	56.15	85.08	69.08	94.59
700	68.26	57.03	87.09	70.09	95.71

TABLE 12. Overall performance comparison (in %).

Parameters	DLBPL	DLASP	EEA	ODWSN	EEOM
P_{th}	72.90	53.02	75.64	64.12	90.44
CSI	72.28	48.41	75.80	63.80	90.33
ΔP	79.29	56.22	82.14	70.96	93.93
MCC	62.54	52.41	75.13	64.87	90.06
FMI	65.92	54.33	81.28	66.81	92.17

1. The incorporation of Machine learning algorithms in this mode of work. The energy optimization problem is complex and as a result, traditional optimization algorithms often perform poorly as compared to their counterparts that incorporate machine learning algorithms. Incorporating machine learning algorithms in this enhanced energy optimization mode of work will help to improve the accuracy and efficiency of the optimization process.

2. Developing data driven models for energy optimization. Real-time data is required to be gathered from the various sources in the energy networks in order to properly understand the energy utilization patterns. Therefore, data-driven models are required to be developed which can effectively make use of the real-time data and optimize the energy utilization in a more efficient way.

3. Robustness of the energy networks is another challenge. The energy optimization process should ensure the robustness of energy networks as they are more prone to cyber-attacks and thus, any kind of ankle in the security protocols can lead to loss of energy.

These are some of the challenges faced in the proposed enhanced energy optimization mode work. For future research, the following can be explored:

1. Developing energy-efficient algorithms for optimization. Machine learning algorithms have so far been more successful in very high-dimensional optimization problem. It can be useful if algorithms are developed which are optimized for the energy networks in order to improve the energy efficiency.

2. Investigate the potential of applying data mining techniques in the energy optimization process. As the amount of data related to energy networks are increasing, the application of data mining algorithms will help to identify useful patterns that can be used to optimize the energy utilization.

3. Research into the application of Blockchain technology in the energy networks. Blockchain technology can improve

TABLE 13. Overall energy consumption (in %).

Parameters	DLBPL		DLASP		EEA		ODWSN		EEOM	
	(C)	(S)	(C)	(S)	(C)	(S)	(C)	(S)	(C)	(S)
TEC	92.27	7.73	76.46	23.54	70.94	29.06	85.08	14.92	64.72	35.28
REC	95.37	4.63	81.30	18.70	72.19	27.81	83.47	16.53	67.27	32.73
IEC	85.32	14.68	59.90	40.10	77.25	22.79	86.43	13.57	52.16	47.84
SEC	93.55	6.45	76.09	23.91	70.36	29.64	80.36	19.64	66.31	33.69

the security of energy networks as it provides an additional layer of encryption. With the use of Blockchain technology, energy networks can become more secure and resilient against any kind of cyber threats.

VII. CONCLUSION

The EEOM can help to improve industrial wireless sensor networks. The EEOM utilizes Machine Learning (ML) to optimize the incorporation of sensor data into industrial operations. It enables an interactive, highly accurate, efficient process and data management. With the EEOM, industrial wireless sensor networks can be configured with low energy and high accuracy, thus reducing the costs associated with installing, maintaining, and operating industrial sensor networks. The EEOM ML model uses evolutionary algorithms to process data from sensors and other nodes in the network. This process manages the power of individual sensors, nodes, and subnets to optimize energy usage and increase accuracy. The proposed EEOM obtained 64.72% transmission energy consumption, 35.28% transmission energy saving, 67.27% received energy consumption, 32.73% received energy storage, 52.16% idle-mode energy consumption, 47.84% idle-mode energy storage, 66.31% sleep-mode energy consumption, and 33.69% sleep-mode energy storage. It also obtained 90.44% prevalence threshold, 90.33% critical success index, 93.93% Delta-P, 90.06% MCC and 92.17% FMI rates. It can also consider different energy sources, such as sunlight and wind, to ensure the most efficient operation. The model can reach more accurate results by including the possible location of individual nodes in its calculations. The EEOM ML model also includes a centralized control unit to oversee the operation of the entire industrial data management system. It ensures maximum accuracy and minimizes the number of inaccurate readings. By leveraging the potential of ML, the EEOM model efficiently learns from errors and calculates the best settings for each node in the network. The EEOM model is highly beneficial for industrial wireless sensor networks. Significant cost savings can be achieved by optimizing the data collection and management process. Additionally, by relying on ML to optimize the network operation, the settings can be tailored to the specific conditions to obtain the best results. The EEOM model is an excellent way for manufacturers and industrial operators to reduce the cost and maintain high accuracy in their wireless networks.

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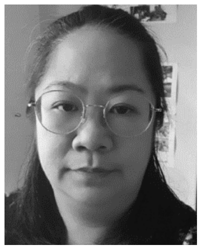

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